Computer Fundamental

STORAGE DEVICE/COMPUTER MEMORY

❖ Secondary Memory / External Memory

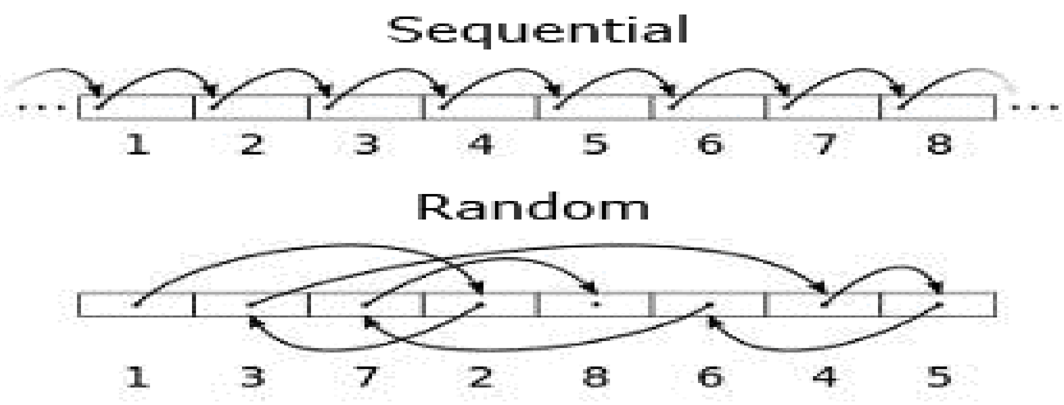
- Which Data is not required by CPU currently (उस समय) then stored in Secondary Memory.
- Store large amounts of data and information .
- It is mainly used to keep backup data,hence it is also called backing storage memory.
- Main memory is temporary and has limited capacity, hence secondary memory is used to store large amount of data and permanent store.

Types of Secondary device based on access

1. **SASD (Sequential Access Storage Device)**
2. **DASD (Direct Access Storage Device)**

1. SASD

- Those devices do not allow the user to access any record directly.
- The data stored on SASD is sequentially.
- If in computer search 30th records, then traverse the previous 29th records and finally show 30th record.
- Its like music cassette (कैसेट) to listen 10th song you will have to forward first to ninth.
- For Example- Magnetic tape (कैसेट)



2. DASD

- This devices that can directly read or write to a specific place.
- The data access on DASD in random way.
- For example –Magnetic Disk, Optical disk etc

Difference b/w SASD & DASD

- Sequential access
  - Sequential access storage devices (SASD)
  - Access in the same order it was written
  - Ex: Need to access memory location 5. Then, you need to go through 1, 2, 3, and 4 first.
- Direct access
  - Direct access storage devices (DASD)
  - Directly access the location
  - Faster than sequential access

Type of Secondary memory

i) Magnetic Disk/ Storage

e.g –Floppy Disk, Hard disk

ii) Optical Disk

e.g – CD, DVD,Blue-Ray

iii)Solid State Disk (SSD) / Flash Memory

e.g – Pen drive, Memory card

i). Magnetic Tape

- A magnetic tape is similar to an audio tape.
- In magnetic tape only one side of the ribbon is used for storing data.
- It is sequential memory which contains thin plastic ribbon to store data and coated by iron oxide / Chromiym die oxide.
- It is used to store large amounts of data.
- Data can be written and erased repeatedly(बार-बार) in it. Magnetic tape drive is a device used for written & erased data in magnetic tape.
- Storage Capacity – 40 Mb to 8GB (a/c to size)

Advantages :

1. These are inexpensive, i.e., low cost memories.
2. It provides backup storage.
3. It can be used for large files.
4. It can be used for copying from disk files.
5. It is a reusable memory.
6. It is compact and easy to store on racks.

Disadvantages :

1. Sequential access is the disadvantage, means it does not allow access randomly or directly.
2. It requires caring to store, i.e., Weak humidity(कमजोर आर्द्रता), dust free, and suitable environment.
3. It stored data cannot be easily updated or modified, i.e., difficult to make updates on data.

i) Magnetic Disk

- Magnetic disks are flat circular plates of metal or plastic, coated on both sides with iron oxide. Input signals, which may be audio, video, or data, are recorded on the surface of a disk as magnetic patterns or spots in spiral tracks & Sectors.
- When data is not require in present (any month/years) then data safely and any damage to used magnetic disk.

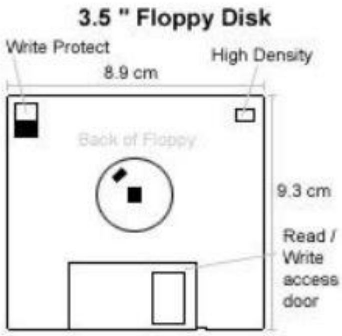
Two common types of magnetic disks are used widely.

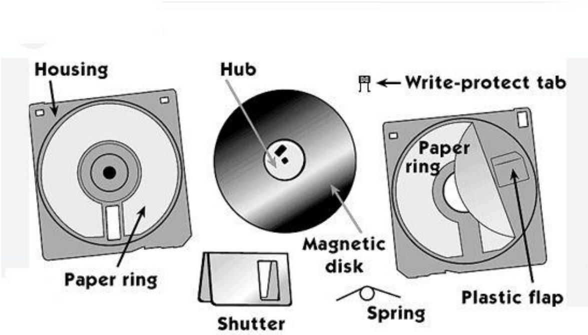
i) Floppy disks **ii) Hard disks**

i) Floppy Disk

- Floppy disks were introduced by IBM in 1971.
- It is also called Diskette(डिस्केट)
- A floppy disk is a round, flat/plane piece of flexible plastic which is coated with magnetic oxide.
- It is covered in square plastic cover that gives protection to the disk.
- The data is read and write in floppy disk is using a device called FLOPPY DISK DRIVE(FDD).
- Floppy Disk available in three size – $3\frac{1}{2}$”, $5\frac{1}{4}$” , 8 inch (8”).

- Capacity of Floppy disk – 3.5” – 1.44 MB, 5.25”-160 kb single side, 8” – 80 KB.
- Floppy disks have tracks and sectors that store data.
- The process of dividing tracks and sectors is called formatting.
- In floppy disk data written in a series of sectors form.





Advantages:

- Cheap in cost.
- Easy offline storage for small computer users.

Disadvantages :

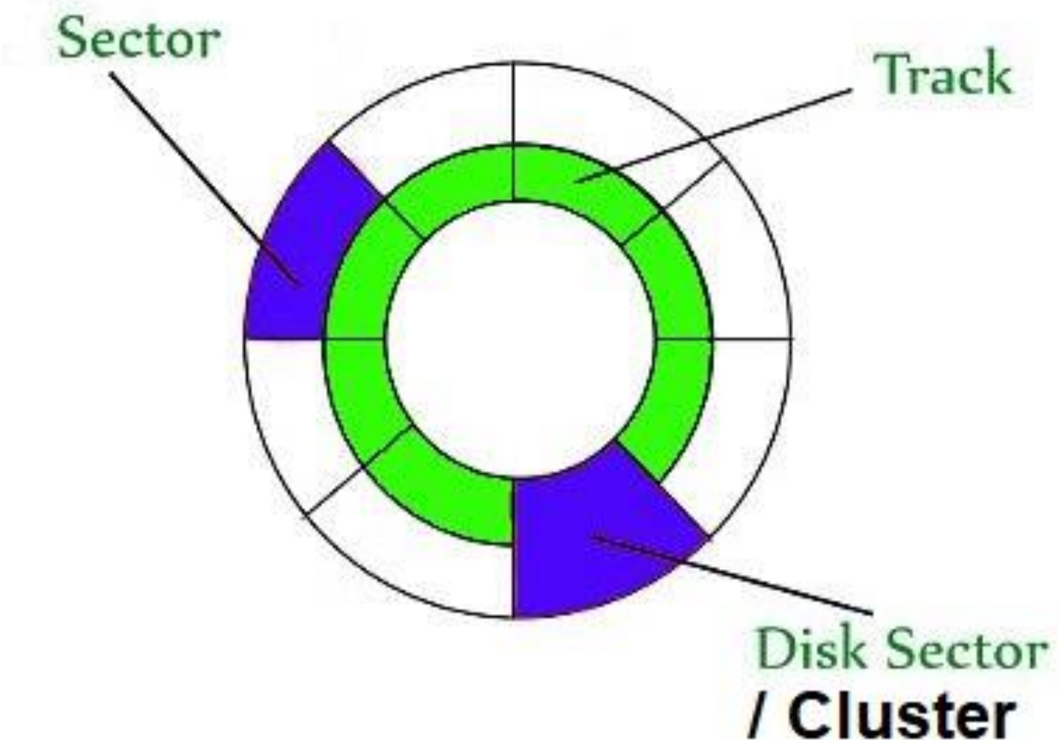
- Low storage capacity.
- A floppy disk drive device is required to use read/write data.

➢ Tracks:

- Tracks are concentric circular paths on the surface of a disk platter.
- Each platter in an HDD is divided into many tracks.
- These tracks serve as the basis for organizing and storing data on the disk.
- Tracks are numbered sequentially, starting from the outer edge and moving inward.
- Multiple platters in an HDD have aligned tracks, forming a cylinder across the platters.

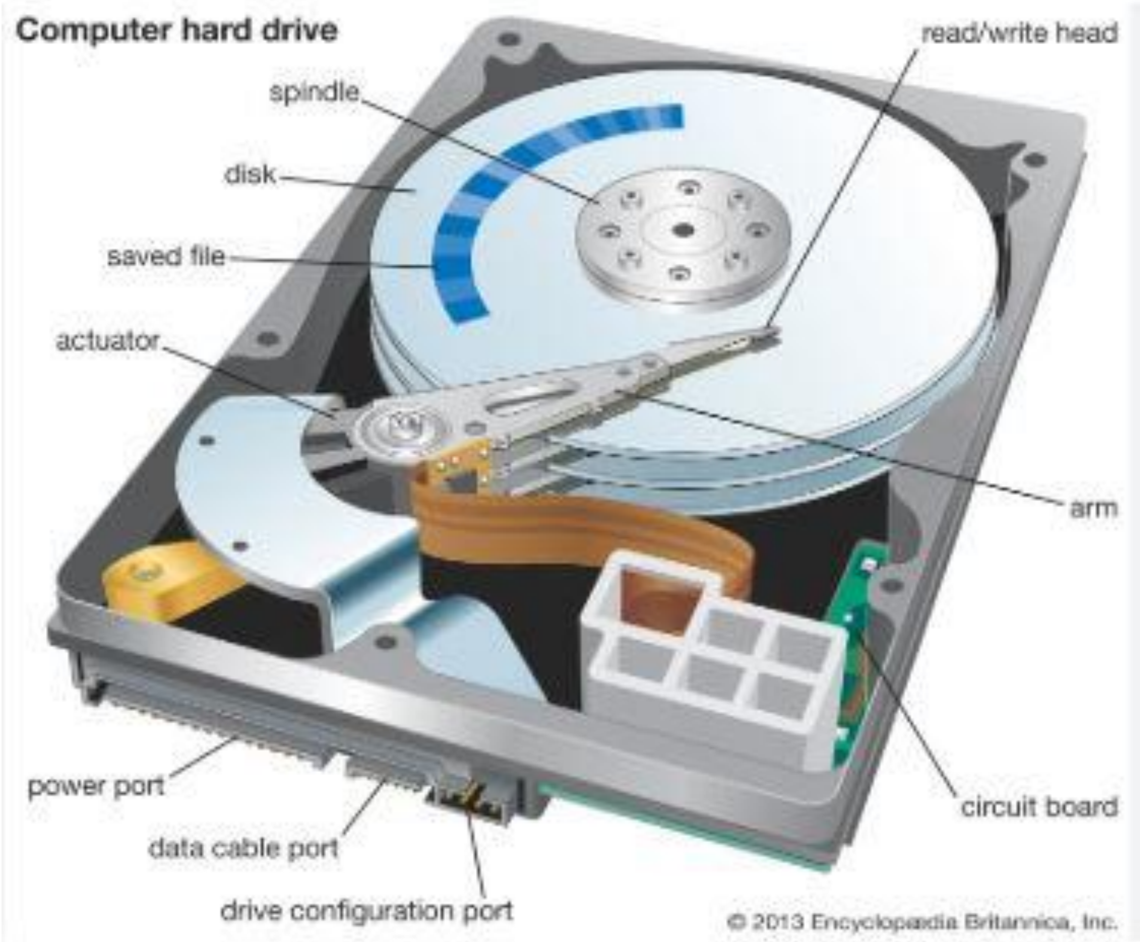
➢ Sectors

- Tracks are further divided into smaller segments called sectors.
- A sector is the smallest unit of data storage on a disk.
- Each sector typically stores 512 bytes or 4 KB of data, depending on the disk format.
- Sectors are identified by a unique address, which includes track and sector numbers.



ii) HARD DISK

- Hard disks are most popular secondary storage device.
- It supports the direct access of the data.
- It is also called Hard drive / Hard disk drive (HDD).
- It's a thin magnetic plate which is made of metal n both side coated with magnetic material.
- The disk is divided in many tracks & Sectors in which data is store in both side of the disk.Mainly store data in Sectors.
- Size of sector store 512/0.5 KB bit Data
- The disk pack consists of multiple disk plates.
- The disk drive pack has a separate read/write head for each disk surface.
- The disk drive consists of motor to rotate the disk pack about its axis at a speed of about 5400 RPM (revolution per minute) to 7200 RPM.
- The arm assembly is capable of moving in & out in direction.
- The hard disk drive has become the most essential secondary storage device in micro-computers.
- It is fast and speeds of less than 10 (ms) milliseconds are achievable the data.
- HDD connected to mother board, Data cable used for connection is PATA, SATA, SCSI.
- PATA – Parallel Advanced Technology Attachment – Old & Slow – 133 MB/S
- SATA – Serial Advanced Technology Attachment – New/Latest & Fast – 150 MB/S – 600 MB/S
- SCSI – Small computer system Interface – 640 MB/S
- Size of HDD – 2.5" & 3.5" (inch)
- HDD maximum size capacity = 20TB
- HDD Manufacturer name – Western Digital – American, Hitachi – Japanese, Seagate – American, Samsung – South Korean.



➢ CD (Compact Disc) / CD - ROM

- Compact disc is backup storage device, which can hold large amounts of data.
- It is type of Optical Disk.
- Usually it is written only once and then it can be read any number of times.
- Laser technology is used to read and write data on these. Hence these are also called optical disc.
- It's a shiny silver colour metal disk of 5 ¼ inch and the storage capacity about 650 MB – 800 MB (megabytes).
- Many of today's micro-computers come with CD-ROM readers and as a result, CD-ROM is popularly used for distribution of software, digitized graphic images as well as Multi-Media material.
- It's known as WORM (Write Once Read Many) disk technology.
- Mainly types of three parts – a) CD-ROM, b) CD-R (Recordable), c) CD-RW (RE-writeable)

Advantages:

- Cost per bit is Low.
- Need not have any mechanical read/write heads to read/write data.
- Compact in size.
- Light weight

Disadvantages:

- Read only storage medium.
- Slower access speed than magnetic disk.

iii) DVD (Digital Video Disc)

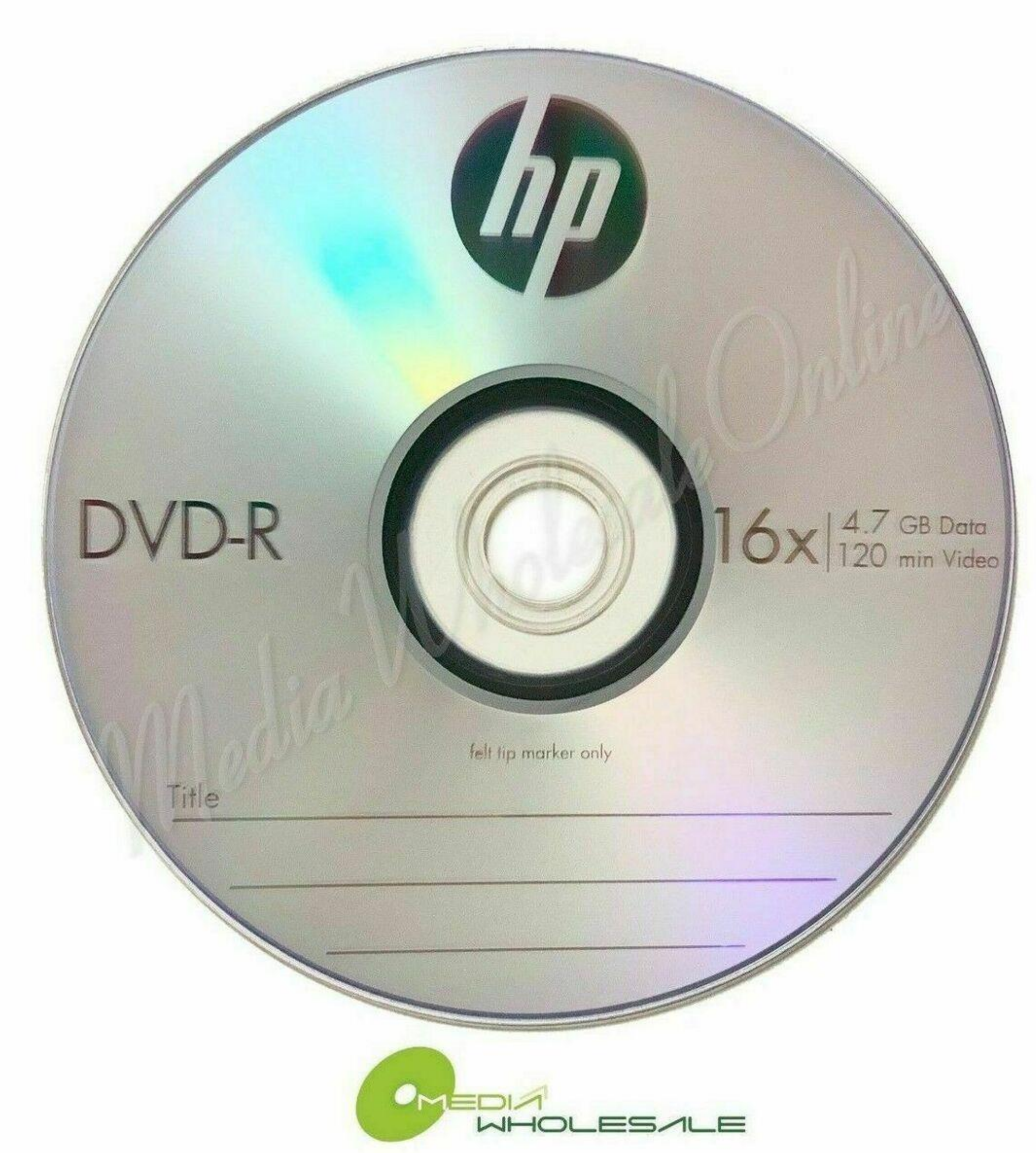
- Basically it is used for storing large amount of data including movies with high video & sound quality.
- It is type of Optical Disk.
- Work on the same principle of CDROM.
- Data is recorded on each layer so that the storage capacity is become large.
- Total capacity of DVD is 8.5 GB

Advantages

- Larger capacity than CD.

Disadvantages:

- Expensive than CD.
- Damaged if not handled properly.



iv) BD (Blue Ray Disc)

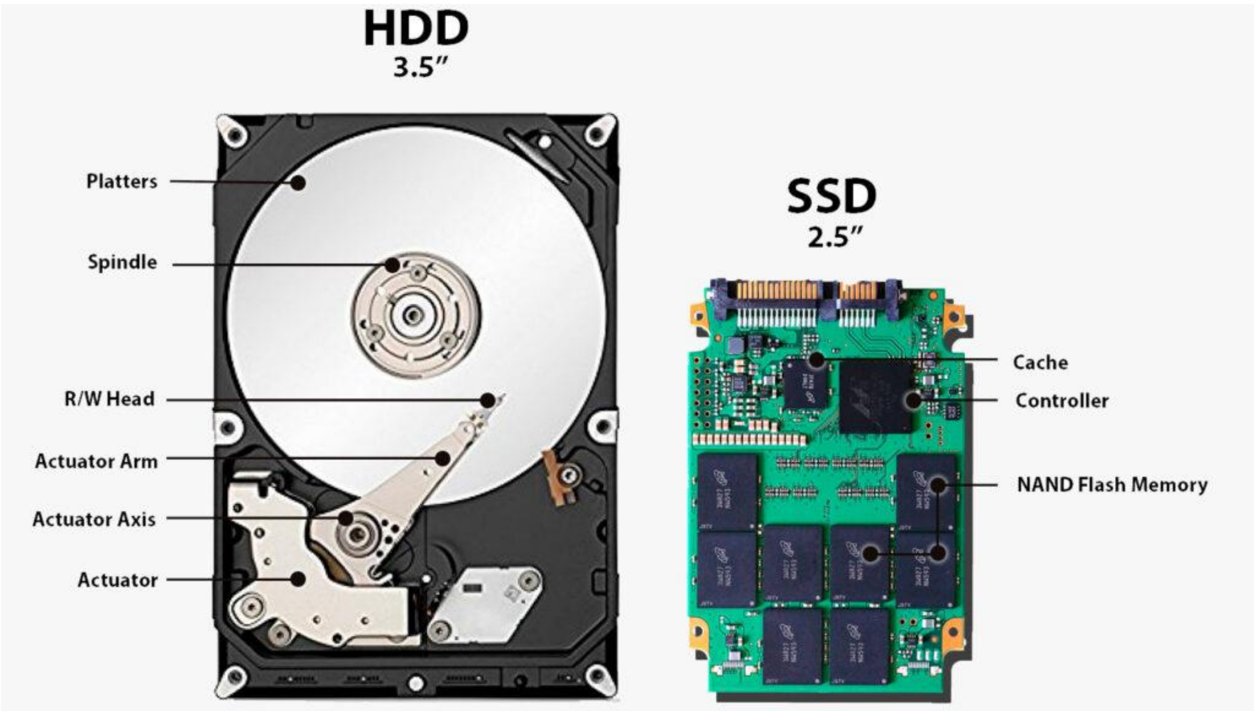
- Blue-Ray disk is an optical disc storage media format.
- It is type of Optical Disk.
- It was developed by blue-ray disc associations.
- It is mainly used to store high definition video and data.
- It has same dimensions as CD or DVD.
- The violet coloured laser is used to read and write the data.
- Because of its shorter wavelength more data can be stored than DVD format.

- Its storage capacity is 50 GB.

v) SSD (Solid State Drive) / Flash Memory

- A SSD is a newer, faster type of device that stores data on instantly-accessible memory chips (Flash Based Memory).
- It’s a Fixed size 2.5” inch solid state drive (SSD).
- It can hold 16TB of data.
- Samsung unveiled (built) the largest capacity SSD in the World.


HDD
CORSAIR
SATA 3
SOLID-STATE DRIVE
240GB
FORCE
HDD Vs SSD


HDD
3.5″
Platters
Spindle
R/W Head
Actuator Arm
Actuator Axis
Actuator
SSD
2.5″
Cache
Controller
NAND Flash Memory

HDD vs SSD: Key Differences

Difference	SSD	HDD
Speed	Faster read/write speeds	Slower read/write speeds
Capacity and Associated Costs	Smaller capacity, higher cost per GB	Larger capacity, lower cost per GB
Operational Reliability	No moving parts, less prone to physical damage	Moving parts, more susceptible to shocks
Power Consumption and Battery Life	Consumes less power, extends battery life	Consumes more power, can reduce battery life
Additional Storage Costs	Can be more expensive for upgrades	Cheaper upgrades and additional storage
Nature of Operations	No mechanical movement	Mechanical movement
Weight and latency	Lighter, lower latency	Heavier, higher latency
Initial Costs and Investment	Higher initial cost	Lower initial cost
Limits on Lifespan	Finite write cycles, potential faster wear	Mechanical wear over time
Security and Data Recovery	Faster secure erasing	Slower secure erasing
Operational Efficiency	Better performance under multitasking	May slow down under heavy multi-tasking

RedSwitches

vi) PEN DRIVE

- It consists of a small printed circuit board encased in a robust plastic or metal casing.
- It is easy to carry in pocket.

- It uses standard-A type connection which allows it to directly connect with the computer.

ADVANTAGES:

- Portable in size.
- Can easily work with all new operating system.
- More reliable than floppy disk

DISADVANTAGES:

- Expensive than optical disk
- Do not provide protect mechanism.

===================HIRA KUMAR=================